5. compiling

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2025-09-12

```r
#Purpose: compile a nice R script using Molly's nice example
#Clear environment
rm(list = ls())

## compiling reports ----
# We will compile reports for homework assignments using a method called
# "spinning." This shows your R code evaluated from start to finish, and shows
# that your results are reproducible.
```

Spinning

When spinning, we can use a different kind of comment, roxygen comments, to do more advanced formatting. I'm switching to roxygen comments now. The formatting only shows up when we compile the document.

We can make text **bold** or *italicized*. There's lots more formatting one can do with markdown, see here: https://www.markdownguide.org/basic-syntax/

You can also create headers and sub-headers for sections.

And, we can write code inline and have it evaluated using backticks. For example: 6.

```r
# This doesn't work in standard comments: `r a + a`
```

You can also write equations, just like in Overleaf (because they both use Latex) For example, $Y = \alpha + \beta_0 + \epsilon$.

You can also write multi-line equations:

$$
\begin{aligned}
\int_0^1 x^2 \, dx &= \frac{1}{3}x^3 \bigg|_0^1 \\
&= \frac{1}{3}1^3 - \frac{1}{3}0^3 \\
&= \frac{1}{3}
\end{aligned}
$$

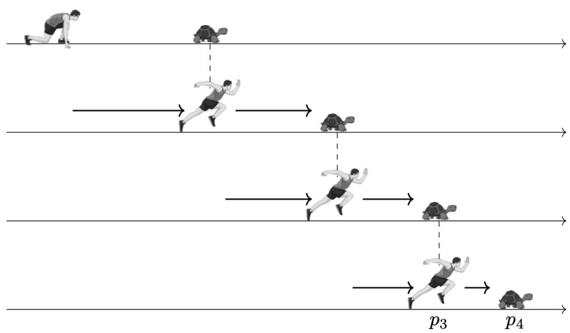
$$
\begin{aligned}
X &= \frac{15+30}{2} - \frac{15+20+20+10+15}{5} \\
&= 6.5
\end{aligned}
$$

A convenient and helpful "cheatsheet" for LaTeX math is here: http://reu.dimacs.rutgers.edu/Symbols.pdf

A tool on the web that might be helpful for writing expressions in a relatively intuitive way is here: http://www.codecogs.com/latex/eqneditor.php?lang=en-en, where you click on symbols and things, and the latex code comes out in the yellow box.

You can also include images, using the `knitr::include_graphics` function.

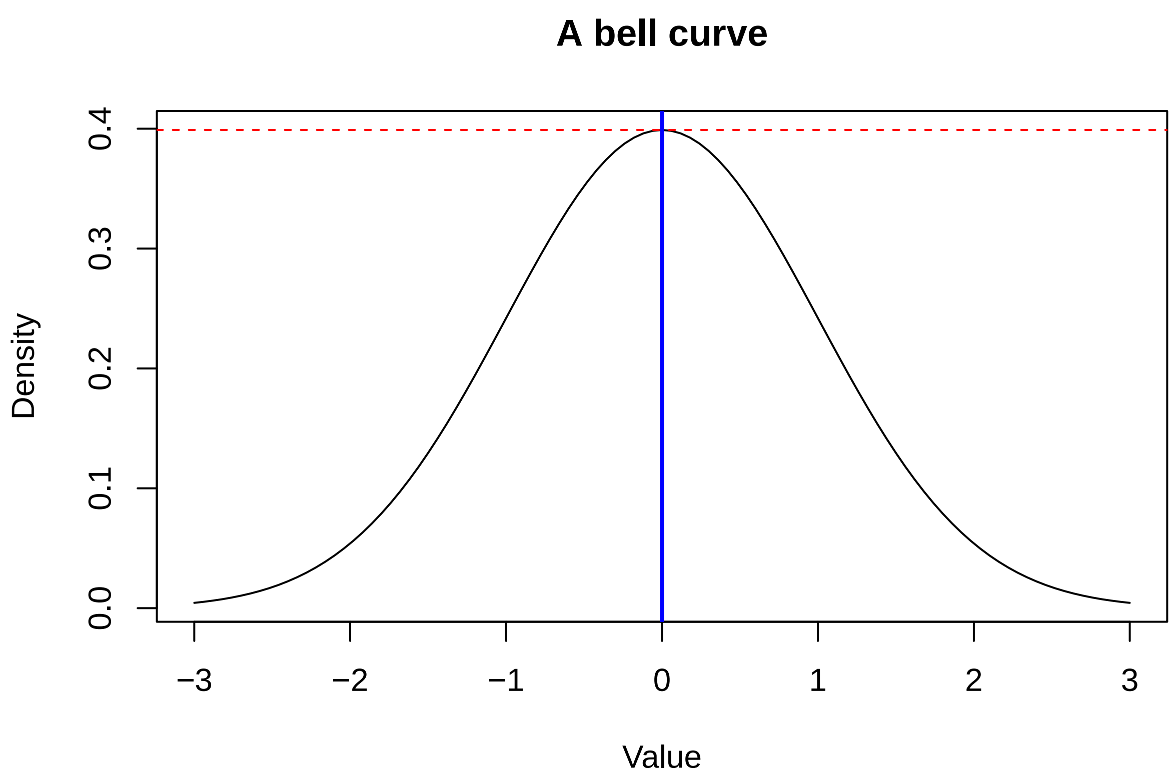
```
knitr::include_graphics("../data/Zenos.jpg")
```



We set the scale of the picture using the out.width argument (here, it's 50%) of the page.

And we can include plots generated in R.

```
plot(dnorm, -3, 3, main = "A bell curve", ylab = "Density", xlab = "Value",
     type="l")
abline(v = 0, lwd = 2, col = "blue") # a vertical line
abline(h = dnorm(0), lty=2, col = "red") # a horizontal dotted line
```



A note on compiling: Reports are compiled using a knitr function called "spin." You can look up `knitr::spin` if you want to learn more options for your text and code presentation. Knitr uses Markdown formatting, which you can look up for things like styling text and creating lists.

Let's compile the document now.

Exercise 1: We hit an error, so let's find and correct it. Now that it's working, what if we want to change where the compiled output lives?